

Fysikk 1    Konstanter

Konstanter

|                                    |                                                       |
|------------------------------------|-------------------------------------------------------|
| Konstanten i Wiens forskyvningslov | $a = 2,90 \cdot 10^{-3} \text{ K m}$                  |
| Boltzmanns konstant                | $k = 1,38 \cdot 10^{-23} \text{ J/K}$                 |
| Stefan–Boltzmann-konstanten        | $\sigma = 5,67 \cdot 10^{-8} \text{ W/m}^2\text{K}^4$ |
| Plancks konstant                   | $h = 6,63 \cdot 10^{-34} \text{ J s}$                 |
| Bohrs konstant                     | $B = 2,18 \cdot 10^{-18} \text{ J}$                   |
| Lysfarten i vakuum                 | $c = 3,00 \cdot 10^8 \text{ m/s}$                     |
| Elementærladningen                 | $e = 1,60 \cdot 10^{-19} \text{ C}$                   |
| Tyngdeakselerasjonen               | $g = 9,81 \text{ m/s}^2$                              |
| Atommasseenheten                   | $u = 1,66 \cdot 10^{-27} \text{ kg}$                  |
| Lysår                              | $1 \text{ ly} = 9,46 \cdot 10^{15} \text{ m}$         |

Masser

|               |                                                                     |
|---------------|---------------------------------------------------------------------|
| Elektron      | $5,4858 \cdot 10^{-4} \text{ u} = 9,1094 \cdot 10^{-31} \text{ kg}$ |
| Proton        | $1,007\,276 \text{ u} = 1,6726 \cdot 10^{-27} \text{ kg}$           |
| Nøytron       | $1,008\,665 \text{ u} = 1,6749 \cdot 10^{-27} \text{ kg}$           |
| Hydrogen      | $1,007\,825 \text{ u}$                                              |
| Deuterium     | $2,014\,102 \text{ u}$                                              |
| Tritium       | $3,016\,049 \text{ u}$                                              |
| Helium-3      | $3,016\,029 \text{ u}$                                              |
| Helium-4      | $4,002\,603 \text{ u}$                                              |
| Alfapartikkel | $4,001\,506 \text{ u}$                                              |

Fysikk 1    Formler

Bevegelse og krefter

|                                               |                                           |            |
|-----------------------------------------------|-------------------------------------------|------------|
| $v = at + v_0$                                | $s = \frac{v + v_0}{2} t$                 |            |
| $v^2 - v_0^2 = 2as$                           | $s = \frac{1}{2}at^2 + v_0t$              |            |
| $\Sigma F = 0 \iff v$ konstant og rettlinjett |                                           |            |
| $\Sigma \vec{F} = m\vec{a}$                   | $F^* = F$                                 | $G = mg$   |
| $R = \mu N$                                   | $B = \rho_f Vg$                           | $L = kv^2$ |
| $\vec{p} = m\vec{v}$                          | $\Sigma \vec{F} \cdot t = \Delta \vec{p}$ |            |
| $m_A v_A + m_B v_B = m_A v_{A0} + m_B v_{B0}$ |                                           |            |

Mekanisk energi og effekt

|                                                                                                                    |                             |                 |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------|
| $W = Fs \cos \theta$                                                                                               | $W_{\Sigma F} = \Delta E_k$ | $P = W/t$       |
| $E_k = \frac{1}{2}mv^2$                                                                                            | $E_p = mgh$                 | $E = E_k + E_p$ |
| $\frac{1}{2}mv^2 + mgh = \frac{1}{2}mv_0^2 + mgh_0$                                                                |                             |                 |
| $\eta = \frac{\text{nyttbar energi}}{\text{tilført energi}} = \frac{\text{nyttbar effekt}}{\text{tilført effekt}}$ |                             |                 |

Trykk og temperatur

|                            |                          |                       |
|----------------------------|--------------------------|-----------------------|
| $p = F/A$                  | $T = t + 273$            | $E_k = \frac{3}{2}kT$ |
| $c = \frac{Q}{m \Delta T}$ | $C = \frac{Q}{\Delta T}$ | $\Delta U = W + Q$    |

Jorda

|                                  |                                  |
|----------------------------------|----------------------------------|
| Masse                            | $5,972 \cdot 10^{24} \text{ kg}$ |
| Middelradius                     | $6371 \text{ km}$                |
| Middelavstand fra sola           | $1,496 \cdot 10^{11} \text{ m}$  |
| Lysintensitet utenfor atmosfæren | $1361 \text{ W/m}^2$             |

Sola

|                     |                                 |
|---------------------|---------------------------------|
| Masse               | $1,99 \cdot 10^{30} \text{ kg}$ |
| Radius              | $6,96 \cdot 10^8 \text{ m}$     |
| Overflatetemperatur | $5780 \text{ K}$                |
| Utstrålt effekt     | $3,83 \cdot 10^{26} \text{ W}$  |

SI-prefikser

|            |       |       |           |       |    |           |      |   |
|------------|-------|-------|-----------|-------|----|-----------|------|---|
| $10^{-18}$ | atto  | a     | $10^{-3}$ | milli | m  | $10^3$    | kilo | k |
| $10^{-15}$ | femto | f     | $10^{-2}$ | centi | c  | $10^6$    | mega | M |
| $10^{-12}$ | piko  | p     | $10^{-1}$ | desi  | d  | $10^9$    | giga | G |
| $10^{-9}$  | nano  | n     | $10^1$    | deka  | da | $10^{12}$ | tera | T |
| $10^{-6}$  | mikro | $\mu$ | $10^2$    | hekto | h  | $10^{15}$ | peta | P |

Trykk og temperatur (forts.)

$$COP = \frac{Q_{\text{avgitt}}}{E_{\text{elektrisk}}}$$

$$COP_{\text{Carnot}} = \frac{T_{\text{avgitt}}}{T_{\text{avgitt}} - T_{\text{opptatt}}}$$

Bølger og stråling

|                                     |                  |                          |
|-------------------------------------|------------------|--------------------------|
| $f = 1/T$                           | $v = \lambda f$  | $U = P/A$                |
| $\lambda_{\text{topp}} \cdot T = a$ | $U = \sigma T^4$ | $I = \frac{P}{4\pi r^2}$ |

Atomfysikk og kjernefysikk

|                               |                                                       |                                   |
|-------------------------------|-------------------------------------------------------|-----------------------------------|
| $E = hf = \frac{hc}{\lambda}$ | $E_n = -\frac{B}{n^2}$                                |                                   |
| $hf = E_n - E_m, \quad n > m$ | $hf = B \left( \frac{1}{m^2} - \frac{1}{n^2} \right)$ |                                   |
| $A = Z + N$                   | $E_0 = mc^2$                                          | $\Delta E_0 = \Delta m \cdot c^2$ |

Elektrisitet

|                                                                                                                |                                 |                   |
|----------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------|
| $I = \frac{Q}{t}$                                                                                              | $U = \frac{W}{Q}$               | $R = \frac{U}{I}$ |
| $W = UI t$                                                                                                     | $P = UI = RI^2 = \frac{U^2}{R}$ |                   |
| Seriekobling: $R_{\text{tot}} = R_1 + R_2 + \dots + R_n$                                                       |                                 |                   |
| Parallellkobling: $R_{\text{tot}} = \left( \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n} \right)^{-1}$ |                                 |                   |